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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	I Forget
Selected Domain	Domain 2 - Orbyte Smart SME Challenge
Selected SDG Track (1–17)	8 - Decent Work and Economic Growth
Project / Solution Name	ORO AI
Name (Leader)	Wong Zhong Jie
Email & Phone (Leader)	jiewongzhong@gmail.com ,017-812-2883
University Affiliation(s)	Asia Pacific University of Technology & Innovation (APU)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small service SMEs such as motorcycle repair and air-conditioning workshops often receive informal customer requests while managing prices, leave and job assignments in separate records. Before promising an appointment, a staff member may need to interpret a message, calculate a quote, check which technicians are available, and confirm who has the necessary skills. These manual cross-checks can be inconsistent or overlooked, especially when a customer expects a quick reply.

The resulting problem is not merely a slow quotation: it is an unreliable customer commitment. A business may accept a date without enough qualified staff, then face rescheduling, potential lost revenue and reputational damage. Later emergency leave can also put an approved booking at risk.

Team I Forget developed this concept through research and observation of technician-based small businesses, including motorcycle repair settings; it has not yet completed formal SME interviews or measured market-wide losses. The target users are Malaysian micro and small service businesses that rely on messages and manual coordination.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

ORO AI is a responsive web prototype that links a customer enquiry with a quotation, technician capacity check and manager decision. In a synthetic Cheras air-conditioning scenario, the message asks for six 1.5HP installations on Friday, 23 October 2026. The catalogue calculates 6 x RM190 = RM1,140. Simulated Orbyte People, eLeave and Work adapters supply staff skills, approved leave and existing jobs. Deterministic scheduling logic finds Friday infeasible and offers Monday, 26 October with a qualified crew. A manager rechecks availability, approves the booking and creates an audit record.

A second demonstration simulates a technician's emergency leave after confirmation. The booking is flagged At Risk, a qualified replacement is proposed, and the manager approves the recovery. The complete workflow is trigger -> connected data -> decision -> approval -> visible outcome.

AI is used where it offers specific value, not for prices or availability: supported browser speech recognition can turn voice into editable text. An external speech-to-text pathway exists in code but is not currently configured. Text enquiry extraction is presently rule-based, not an active large language model; more flexible LLM parsing is a future enhancement. Development used AI coding assistance with human review. Workshop-related methods used include database architecture, API design and Replit cloud deployment. All Orbyte integration and the credit wallet are simulated; no production platform is connected.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 19, TypeScript, Vite 7, Tailwind CSS 4; responsive/mobile PWA provisions. Backend: Node.js 24, Express 5, TypeScript; validated HTTP endpoints using OpenAPI and Zod. Data: SQLite (synthetic staff, leave/job capacity blocks, quotations and activity log). Hosting: Replit deployment. Source control: GitHub (currently private; public access for judges is pending). Testing: Node test runner; latest Replit workspace audit reported 44 passing tests, 0 failures.

Architecture:

Customer text / supported browser voice

-> enquiry parser -> catalogued quotation

-> mock Orbyte People/eLeave/Work data

-> capacity and conflict engine

-> manager role gate + database recheck

-> confirmed booking + activity log

-> simulated leave-risk and recovery flow.

Sponsor-provided platform: Replit Core ($20 plan) is the compulsory sponsor resource for this hackathon. Our team uses Replit for application development and hosting.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We began with a narrow pain point relevant to technician-based SMEs: verifying capacity before promising a service date. To define user needs we mapped the steps an owner would take from a WhatsApp-style

enquiry to a confirmed job; this is preliminary research and process modelling, not a completed customer interview study.

We prototyped one end-to-end story in Replit using fictional staff, services, leave and bookings. We prioritised a readable dashboard, conflict explanations, manager approval and an activity log so non-technical users can understand each recommendation. The quote and scheduling engines use testable rules rather than guesses; simulated Orbyte connectors separate business data from decision logic.

The current Replit workspace reports 44 passing automated checks, including competing approvals and the recovery scenario. Phase 2 priorities are deployment-access verification, browser walkthrough testing, fixing stale analyses after enquiry edits, stronger data persistence and trial feedback from SME staff. More flexible AI extraction will be considered only after the core workflow is reliable.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.2: productivity through technological upgrading and innovation. SDG 8.3: entrepreneurship and growth of micro, small and medium-sized enterprises. ORO AI's direct contribution is a testable workflow that helps service SMEs cross-check technician capacity before approval.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	According to the OECD (2024), only **54% of Malaysian micro, small and medium enterprises (MSMEs) used back-end digital processes**, based on earlier SME digitalisation research. Furthermore, Khazanah Research Institute (2026) reported that digital adoption among Malaysian micro and small enterprises remains concentrated in customer-facing activities, while internal business systems are often fragmented and poorly integrated. These findings indicate an ongoing need for more coordinated digital solutions that support SME operations.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	PRIMARY KPI: Conflict-detection accuracy (%) = infeasible appointment requests correctly blocked / all known-infeasible test requests x 100.

Metric Category	Your Response
	Evaluate on at least 20 pre-labelled synthetic scheduling cases, with separate checks for leave, existing bookings and crew skills. SECONDARY KPI: Median time (seconds) to reach a confirmed feasible appointment in a manual versus ORO AI controlled walkthrough. Record both results; do not assume savings without timing the tasks.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	- Block at least 95% of genuinely infeasible appointment requests across >=20 labelled synthetic cases. - Require manager approval for 100% of booking confirmations. - Achieve at least 25% lower median end-to-end completion time than a timed, equivalent manual workflow under controlled tasks.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React / Vite / Tailwind CSS / Radix UI	Code — Frontend frameworks, styling and UI components	MIT
Node.js / Express / Zod	Code — JavaScript runtime, backend framework and validation	MIT
TypeScript / Drizzle ORM	Code — TypeScript tooling and database ORM dependency	Apache-2.0
TanStack React Query / Framer Motion	Code — API data management and UI animation	MIT
Lucide React	Code — Interface icons	ISC

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

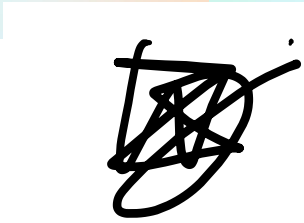
☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

24-01-2026